

CHRISTOPHER CRILLY PIENAABH

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github.com/CrillyPienaah | chris-pienaah-portfolio.vercel.app

TECHNICAL SKILLS

Languages & Databases: Python, SQL, R, JavaScript (Node.js, TypeScript), PostgreSQL, Supabase, DynamoDB, Snowflake, BigQuery
ML / NLP / LLMs: LangChain, LangGraph, Flowise (low-code AI workflows), GPT-4, Claude API, Gemini, Hugging Face, RAG, Transformer-based NLP, Multi-Agent Orchestration, Model Fine-Tuning, SHAP, Prompt Engineering, Embeddings
Integrations & Platforms: Salesforce API, REST/GraphQL APIs, Webhook Automation, CRM Integration, OpenAI API, Supabase, Node.js, TypeScript
ML Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, XGBoost, spaCy, NLTK, Spark NLP
Cloud & MLOps: AWS (S3, Lambda, SageMaker), GCP (Vertex AI, BigQuery ML, Looker), Azure (Data Factory, Databricks), Docker, CI/CD, MLflow, Airflow, Feature Drift Detection, Model Registry, dbt, Power BI
Model Governance: OSFI E-23, SR 11-7, Independent Model Validation, Conceptual Soundness, Uncertainty Quantification, Fairness Auditing, Metric Stability, Data Drift Verification, Explainable AI (SHAP, LIME), Regulatory Traceability

TECHNICAL PROJECTS

- SME Growth Co-Pilot – Multi-Agent AI System** | *Python, LangGraph, GPT-4, Streamlit, PostgreSQL* 2024
- Built a production 6-agent LangGraph orchestration system for the Kaggle × Google AI Agents Intensive capstone – performing competitive intelligence, financial forecasting, and strategic recommendation synthesis with deterministic output validation and 100% reproducible pipeline execution.
 - Cut manual SME analysis time 90% by automating data ingestion, multi-source cross-referencing, and structured report generation across financial and market datasets.
- Self-Healing ML Pipeline & Monitoring Platform** | *GCP, Python, Airflow, MLflow, Docker* 2024–2025
- Reduced post-deployment system downtime 78% by engineering an automated self-healing ML pipeline processing 431M+ behavioral records with drift detection, rollback gates, and model registry versioning across GCP and AWS.
 - Achieved full audit-trail reproducibility across every deployment by building CI/CD-gated retraining workflows with environment isolation, metric stability tracking, and data drift verification aligned to SR 11-7 model risk standards.
- LuminaMed AI – Clinical Radiology Intelligence** | *Python, PyTorch, LangChain, GPT-4, FastAPI* 2024
- Reduced LLM hallucination rate from 12% to 8% by designing a 3-agent RAG verification system benchmarked across 3 architectures; integrated SHAP/LIME explainability layers to satisfy model risk governance and conceptual soundness protocols.
 - Ensured full HIPAA compliance by designing de-identification pipelines and controlled access protocols with MLflow model versioning and reproducible evaluation across every deployment iteration.
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- Nytia Labs – Behavioral Wellness Data Platform** | *GCP, Python, BigQuery, Airflow, dbt* 2024
- Achieved ROC AUC 94% on a behavioral health classification model by building end-to-end ETL pipelines processing 431M+ daily records with automated data quality validation and anomaly remediation workflows.
 - Reduced model governance failures 100% by designing FOIP-compliant access controls, metadata standards, and data stewardship policies audited across the full data lifecycle.
 - Accelerated data science cycle time 40% by engineering a reusable feature store and standardized schema registry enabling reproducible, multi-team access to curated wellness datasets.
- Job Copilot – AI Resume Intelligence Platform** | *FastAPI, React, TypeScript, OpenAI, Flowise* 2025
- Engineered a full-stack AI-powered resume optimization platform using FastAPI, React/Vite, and OpenAI GPT-4o-mini, delivering recruiter-grade match scoring across skills (30%), responsibilities (30%), and experience relevance (40%) with structured JSON output validation.

- Upgraded from keyword matching to semantic AI scoring by integrating OpenAI embeddings and cosine similarity, enabling ATS-style vector matching – mirroring real enterprise screening pipelines.
- Built end-to-end agentic job search automation in Flowise including a resume optimizer chain, job parser agent, and alert emailer – connecting LLM workflows with automated cover letter generation and job fit analysis.

Power BI Analytics Portfolio <i>Power BI, DAX, Azure Maps, World Bank WDI, Python</i>	2025–2026
• Built a 3-page AI Model Monitoring dashboard for Nytia Labs surfacing predicted risk tiers (LOW/MODERATE/HIGH/CRITICAL) across 52K+ flagged patients, connecting Power BI directly to live ML model outputs for operational decision-making. • Delivered 3 additional enterprise dashboards spanning 827K+ accident records, US\$46M+ federal contract opportunities, and Ghana macroeconomic indicators (2015–2024) featuring DAX-driven measures, Azure Maps, and executive-level time-series analysis.	
Dockett – Legal Billing SaaS <i>Node.js, PostgreSQL, Supabase, TypeScript, OpenAI API</i>	2024–2025
• Shipped a production legal SaaS for Ontario independent legal professionals by architecting a secure PostgreSQL schema with role-based access control, compliance audit logging, and OpenAI-powered billing automation – replacing manual invoicing workflows end-to-end.	

PROFESSIONAL EXPERIENCE

AI Engineer & Product Strategist	Sept 2024 – May 2025
<i>ICON Leadership Institute</i>	<i>Toronto, ON</i>
• Increased NLP pipeline throughput 3× (450K → 1.35M daily records) by reengineering GPT-4 workflows with async processing, optimized prompt templates, and Redis caching on AWS Lambda – reducing average inference latency by 55%. • Cut manual data triage duration 60% by designing a high-throughput document classification and entity extraction pipeline using Hugging Face Transformers with structured confidence-scoring gates routing high-risk cases to human validation. • Cut average program startup time 25% by designing a multi-agent orchestration framework integrating regulatory document analysis, participant feasibility scoring, and cross-program benchmarking into a unified AI-assisted workflow. • Eliminated \$15K–\$25K in annual SaaS licensing costs by engineering a custom full-stack CRM from scratch serving 10–20 staff with native integrations to Salesforce and third-party platforms, covering contact management, pipeline tracking, billing automation, and client communications.	
Director of Operations & Analytics	Jan 2014 – Aug 2024
<i>CHRISLINE Prints & Advertising</i>	<i>Ghana / Canada</i>
• Engineered a centralized PostgreSQL database and Python/SQL ETL pipeline automating financial reporting and KPI tracking across 50+ SME client accounts, eliminating manual compilation and reducing turnaround time 65%. • Standardized client-facing dashboard schemas with automated data ingestion validation and anomaly flagging, ensuring data quality and reducing reporting errors to near-zero across all accounts.	

EDUCATION

Northeastern University	Toronto, ON
<i>MPS in Analytics – Applied Machine Intelligence GPA: 3.89/4.0 (Analytics: 3.97/4.0)</i>	<i>May 2026</i>
• Relevant coursework: Machine Learning, NLP, Deep Learning, AI Systems, Data Mining, Applied Mathematics, Reinforcement Learning	
Sir Sandford Fleming College	Toronto, ON
<i>Graduate Certificate in Project Management</i>	<i>July 2024</i>
University of Nottingham (Russell Group)	United Kingdom
<i>Master of Science in Entrepreneurship, Innovation & Management</i>	<i>December 2015</i>
University for Development Studies	Ghana
<i>Bachelor of Arts in Economics & Entrepreneurship Studies</i>	<i>December 2013</i>

CERTIFICATIONS

5-Day AI Agents Intensive with Google – Kaggle × Google (Dec 2025) | Verified badge:
 kaggle.com/certification/badges/christophercrilly/105
AWS Generative AI – Udacity (2025)
Applied AI Development – DTTP (2025)
Google Data Analytics Professional Certificate – Google (2024)